

Coppervale Semiconductor Fab

Case Study 39 - Network Outage Investigation - Cisco Packet Tracer Lab Guide

1. What is this project?

Coppervale Semiconductor has three sites: a **Head Office (HQ)**, a **Fab Plant** (legacy EIGRP) and a **Test & Packaging** site. They are joined by an OSPF backbone and the HQ has dual-ISP Internet using BGP. For 48 hours users have reported random problems. You are the on-call network engineer. In the lab you **build the network, recreate the 5 faults (A to E), diagnose each with show commands, fix it, and prove the fix**, then write a short report.

Work flow for every fault: Reproduce (break it) -> Observe (show commands = BEFORE) -> Explain (why) -> Fix -> Verify (show commands = AFTER).

Sym.	Where	What users see	Real cause in one line
A	Fab access switch	Port blocked, log says root-inconsistent	Neighbor switch sent a better (lower priority) BPDU; Root Guard blocked the port
B	WAN OSPF segment	One neighbor stuck in ATTEMPT	Non-broadcast (NBMA) network: neighbor is not configured/answering correctly
C	Fab to Test&Pkg EIGRP	Expected route missing on other routers	Router configured as EIGRP stub of the wrong type (receive-only)
D	HQ Internet edge (BGP)	eBGP to secondary ISP keeps resetting	ISP sends more prefixes than the maximum-prefix limit, so the session is torn down
E	Fab access layer capture	Burst of large fragmented ICMP echo to one host	Ping flood / DoS attempt

2. Topology and what each device means

This is the final lab topology. Build exactly this. The letters A to E show where each fault lives.

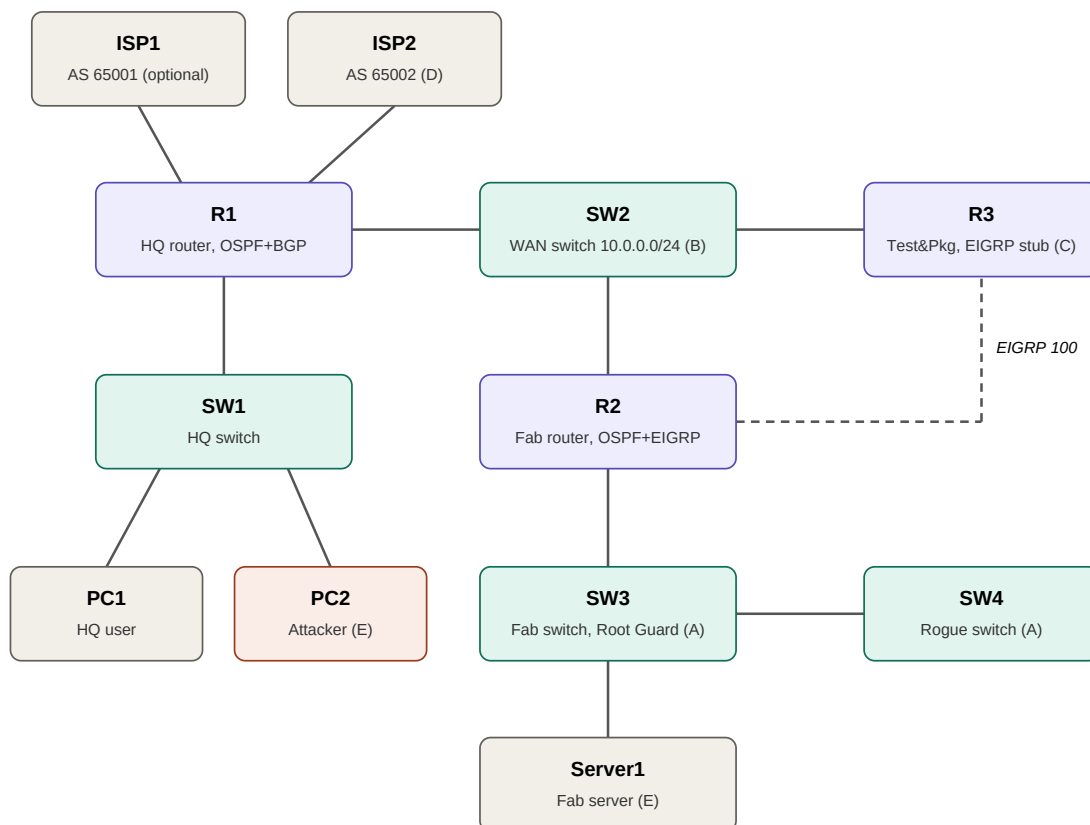


Figure 1 - Lab topology. Solid line = Ethernet cable. Dashed line = EIGRP link between R2 and R3. ISP1 is optional (see note below).

2.1 How to read the names (PC1, SW1, R1...)

- **PC1, PC2** = PC number 1, PC number 2. End-user computers (clients). They only have an IP address and a default gateway.
- **SW1, SW2** = Switch number 1, 2 (Cisco 2960). A switch works at Layer 2: it connects devices inside one network using MAC addresses. It does not route between networks.
- **R1, R2, R3** = Router number 1, 2, 3. A router works at Layer 3: it connects different networks using IP addresses and routing protocols (OSPF, EIGRP, BGP).
- **ISP1, ISP2** = Internet Service Provider routers, outside your company, in their own BGP Autonomous System (AS).
- **Server1** = an internal server used as the target of the ping flood.
- **fa0/1, gi0/0** = interface names. fa = FastEthernet (100 Mbps), gi = GigabitEthernet (1 Gbps); the numbers are slot/port. **lo0** = loopback, a virtual interface that is always up.

2.2 Device table

Name	Device (Packet Tracer)	Role in this project	Addresses
R1 (R-HQ)	Router 2911	HQ router. OSPF area 0 toward the WAN, eBGP to ISP2 (AS 65100). Place of symptom D and OSPF hub for B.	gi0/0 192.168.10.1/24 (to SW1) gi0/1 10.0.0.1/24 (to SW2) gi0/2 172.16.2.1/30 (to ISP2)
R2 (R-FAB)	Router 2911	Fab Plant router. OSPF toward WAN, EIGRP toward R3, redistributes between them.	gi0/0 10.0.0.2/24 (to SW2) gi0/1 192.168.20.1/24 (to SW3) gi0/2 10.2.0.1/30 (to R3)
R3 (R-TP)	Router 2911	Test & Packaging router. EIGRP stub (symptom C). Also on the WAN switch.	gi0/0 10.0.0.3/24 (to SW2) gi0/1 10.2.0.2/30 (to R2) lo0 192.168.30.1/24 (LAN)
ISP2	Router 2911	Secondary ISP, AS 65002. Advertises 5 test networks (198.51.100.0/24 to 198.51.104.0/24) using loopbacks.	gi0/0 172.16.2.2/30 lo1 to lo5
ISP1	Router (optional)	Primary ISP, AS 65001. Only for dual-homing look; skip if your router has no free port.	172.16.1.0/30
SW1	Switch 2960	HQ LAN switch.	fa0/1 R1, fa0/2 PC1, fa0/3 PC2
SW2	Switch 2960	The multipoint WAN segment where R1, R2, R3 meet (symptom B). Plain switch, default config.	fa0/1 R1, fa0/2 R2, fa0/3 R3
SW3	Switch 2960	Fab access layer switch and intended STP root. Root Guard on fa0/3 (symptom A).	fa0/1 R2, fa0/2 Server1, fa0/3 SW4
SW4	Switch 2960	Neighbor switch whose priority was mistakenly lowered (superior BPDU).	fa0/1 to SW3 fa0/3
PC1	PC	Normal HQ user, 192.168.10.10/24, GW 192.168.10.1	-
PC2	PC	Attacker / infected PC, 192.168.10.66/24, GW 192.168.10.1. Sends the ping flood (E).	-
Server1	Server	Fab internal host 192.168.20.100/24, GW 192.168.20.1. Target of the flood.	-

Port note: a 2911 has only 3 gigabit ports, so R1 uses gi0/0, gi0/1, gi0/2 and ISP1 is left out (or add a module). The ISP1 box only shows the dual-ISP design; symptom D is tested on ISP2, the secondary ISP, exactly as in the case study.

2.3 The protocols in one line each

- **STP (Spanning Tree):** stops Layer 2 loops by electing one root bridge (lowest priority wins) and blocking redundant ports. **Root Guard** says: this port must never lead to the root.
- **OSPF:** link-state routing protocol used on the WAN backbone. Neighbors must reach state FULL. On a non-broadcast (NBMA) network there is no multicast, so neighbors must be defined by hand.

- **EIGRP:** Cisco routing protocol used at the Fab Plant. A **stub** router tells neighbors it is not a transit path and only advertises certain route types.
- **BGP:** routing protocol of the Internet. **eBGP** is BGP between different AS numbers (company to ISP). **maximum-prefix** caps how many routes a neighbor may send.
- **ICMP echo:** the 'ping' packet. Used in a flood to overload a host or link.

3. Build steps (Packet Tracer)

Step 1 - Place devices and cables

- Add 3 routers (R1, R2, R3) and ISP2 as **2911**. Add 4 switches **2960** (SW1 to SW4), 2 PCs and 1 Server.
- Use **copper straight-through** for router-to-switch and switch-to-PC. Use **copper cross-over** for switch-to-switch (SW3-SW4), router-to-router (R2-R3, R1-ISP2).
- Connect exactly like Figure 1 and the device table. Rename each device (double click, Config tab, Display name).
- Turn each router on/off only if you add modules; otherwise leave default.

Step 2 - Basic IP configuration (all routers)

Open the CLI tab of each router. Example for R1:

```
Router> enable
Router# configure terminal
Router(config)# hostname R1
R1(config)# interface gi0/0
R1(config-if)# ip address 192.168.10.1 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# interface gi0/1
R1(config-if)# ip address 10.0.0.1 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# interface gi0/2
R1(config-if)# ip address 172.16.2.1 255.255.255.252
R1(config-if)# no shutdown
```

Do the same for R2 and R3 with the addresses from the device table. On R3 also create the LAN as a loopback: interface lo0 then ip address 192.168.30.1 255.255.255.0. Set PC and Server IPs in Desktop -> IP Configuration.

Step 3 - Healthy base configuration (this is your 'working' state)

OSPF (R1, R2, R3)

```
R1(config)# interface gi0/1
R1(config-if)# ip ospf network non-broadcast
R1(config-if)# ip ospf priority 255
R1(config)# router ospf 1
R1(config-router)# router-id 1.1.1.1
R1(config-router)# network 10.0.0.0 0.0.0.255 area 0
R1(config-router)# network 192.168.10.0 0.0.0.255 area 0
R1(config-router)# neighbor 10.0.0.2
R1(config-router)# neighbor 10.0.0.3
```

```
R2(config)# interface gi0/0
R2(config-if)# ip ospf network non-broadcast
R2(config-if)# ip ospf priority 0
R2(config)# router ospf 1
R2(config-router)# router-id 2.2.2.2
R2(config-router)# network 10.0.0.0 0.0.0.255 area 0
R2(config-router)# network 192.168.20.0 0.0.0.255 area 0
```

```
R3(config)# interface gi0/0
R3(config-if)# ip ospf network non-broadcast
R3(config-if)# ip ospf priority 0
R3(config)# router ospf 1
R3(config-router)# router-id 3.3.3.3
R3(config-router)# network 10.0.0.0 0.0.0.255 area 0
```

R1 is the hub (priority 255) and holds the neighbor statements; R2 and R3 are spokes (priority 0).

EIGRP (R2 and R3) and redistribution (R2)

```
R2(config)# router eigrp 100
R2(config-router)# network 10.2.0.0 0.0.0.3
R2(config-router)# redistribute ospf 1 metric 10000 100 255 1 1500
R2(config-router)# exit
R2(config)# router ospf 1
R2(config-router)# redistribute eigrp 100 subnets
```

```
R3(config)# router eigrp 100
R3(config-router)# network 10.2.0.0 0.0.0.3
R3(config-router)# network 192.168.30.0
R3(config-router)# eigrp stub connected summary
```

BGP (R1 and ISP2)

```
ISP2(config)# interface lo1
ISP2(config-if)# ip address 198.51.100.1 255.255.255.0
... repeat for lo2 198.51.101.1, lo3 198.51.102.1, lo4 198.51.103.1, lo5 198.51.104.1
ISP2(config)# router bgp 65002
ISP2(config-router)# neighbor 172.16.2.1 remote-as 65100
ISP2(config-router)# network 198.51.100.0 mask 255.255.255.0
... same for 101.0, 102.0, 103.0, 104.0
```

```
R1(config)# router bgp 65100
R1(config-router)# neighbor 172.16.2.2 remote-as 65002
R1(config-router)# neighbor 172.16.2.2 maximum-prefix 10
```

Spanning tree (SW3 and SW4)

```
SW3(config)# spanning-tree vlan 1 priority 4096      ! SW3 is the intended root
SW3(config)# interface fa0/3
SW3(config-if)# spanning-tree guard root
SW4(config)# spanning-tree vlan 1 priority 32768     ! normal value
```

Step 4 - Check that everything works first

```
R1# show ip ospf neighbor      -> R2 and R3 are FULL
R1# show ip route              -> 192.168.20.0, 192.168.30.0 present (0 E2 or 0)
R1# show ip bgp summary        -> ISP2 shows 5 prefixes
SW3# show spanning-tree inconsistentports  -> none
PC1> ping 192.168.20.100      -> replies
PC1> ping 192.168.30.1       -> replies
```

Take a screenshot or copy this output. This is your **healthy baseline**.

4. Reproduce, diagnose, fix and verify each symptom

For each symptom: run the BREAK commands, capture the BEFORE output, run the FIX, capture the AFTER output. Outputs shown are **representative**; Packet Tracer wording may differ slightly.

Symptom A - Root Guard (Fab access layer)

What Root Guard prevents: an unauthorized or misconfigured switch becoming the root bridge. The root decides the whole Layer 2 tree, so a new root would change traffic paths and could cause outages or let an attacker see traffic. **Why root-inconsistent happens:** the root-guard port receives a *superior BPDU* (lower bridge priority/ID than the current root). Instead of accepting it, the switch puts the port into a blocking state called root-inconsistent. It recovers by itself when superior BPDUs stop.

Break

```
SW4(config)# spanning-tree vlan 1 priority 0
```

Diagnose (BEFORE)

```
SW3# show spanning-tree inconsistentports
Name                Interface          Inconsistency
-----
VLAN0001            FastEthernet0/3      Root Inconsistent
Number of inconsistent ports (segment) in the system : 1
```

```
%SPANTREE-2-ROOTGUARD_BLOCK: Root guard blocking port FastEthernet0/3 on VLAN0001.
```

```
SW3# show spanning-tree interface fa0/3 detail    (look at: Root guard is enabled / port is in the root-inconsistent)
SW4# show spanning-tree vlan 1                    (shows SW4 claims "This bridge is the root", priority 0)
```

Fix and verify (AFTER)

```
SW4(config)# spanning-tree vlan 1 priority 32768
```

```
SW3# show spanning-tree inconsistentports
Number of inconsistent ports (segment) in the system : 0
%SPANTREE-2-ROOTGUARD_UNBLOCK: Root guard unblocking port FastEthernet0/3 on VLAN0001.
SW3# show spanning-tree vlan 1 -> "This bridge is the root", fa0/3 is Desg FWD
```

If the port does not recover, shut/no shut fa0/3 on SW3. Do **not** remove Root Guard; it is the protection. Fix the cause (the priority).

Symptom B - OSPF neighbor stuck in ATTEMPT (multipoint WAN)

Why it happens: on a non-broadcast (NBMA) segment OSPF cannot find neighbors with multicast, so the hub sends unicast hellos to each neighbor address you configure. State **ATTEMPT** means: 'I am sending hellos to this configured neighbor but I have heard nothing back.' So one remote router stays in ATTEMPT while other neighbors on the same segment are FULL when that router's side is not set up to answer (for example it is not configured as non-broadcast, or the hub-side/neighbor statement for it is wrong or missing). If the hub has **no** neighbor statement for a spoke, that spoke never appears at all. Both sides need correct NBMA settings.

Break

```
R2(config)# interface gi0/0
R2(config-if)# no ip ospf network non-broadcast ! back to default (broadcast type)
R2(config-if)# no ip ospf priority 0
```

Diagnose (BEFORE)

```
R1# show ip ospf neighbor
Neighbor ID    Pri  State           Dead Time   Address      Interface
3.3.3.3        0    FULL/DROTHER    00:01:41    10.0.0.3     GigabitEthernet0/1
0.0.0.0        0    ATTEMPT/DROTHER 00:01:52    10.0.0.2     GigabitEthernet0/1

R1# show ip ospf interface gi0/1 -> Network Type NON_BROADCAST, Hello 30
R2# show ip ospf interface gi0/0 -> Network Type BROADCAST, Hello 10 (mismatch found)
R1# show running-config | section router ospf -> check the neighbor lines
```

Fix and verify (AFTER)

```
R2(config)# interface gi0/0
R2(config-if)# ip ospf network non-broadcast
R2(config-if)# ip ospf priority 0
R1(config-router)# neighbor 10.0.0.2 ! add if it was missing on the hub
```

```
R1# show ip ospf neighbor
Neighbor ID    Pri  State           Dead Time   Address      Interface
2.2.2.2        0    FULL/DROTHER    00:01:50    10.0.0.2     GigabitEthernet0/1
3.3.3.3        0    FULL/DROTHER    00:01:41    10.0.0.3     GigabitEthernet0/1
```

Packet Tracer sometimes shows **Down** or **Init** instead of ATTEMPT. If so, screenshot what you see and explain in the report that on a real router the state is ATTEMPT. Use the 'missing neighbor statement' version too: remove neighbor 10.0.0.2 from R1 and show that R2 never appears, then add it back.

Symptom C - EIGRP stub is not advertising a route

Stub types:

Stub option	What the stub advertises	Use
eigrp stub (default)	connected + summary	Normal branch router
eigrp stub connected	only connected networks	Branch with LANs only
eigrp stub summary	only summary routes	Branch with summarized LANs
eigrp stub static / redistributed	static / redistributed routes	Branch with static or redistributed routes
eigrp stub receive-only	NOTHING. It only listens.	Isolated router that must not share routes

Why a wrong type hides a route: a stub does not pass queries and only advertises what its type allows. With *receive-only* the router learns routes but advertises none, even its own connected LAN, so the route exists in its table but nobody else learns it.

Break

```
R3(config)# router eigrp 100
R3(config-router)# eigrp stub receive-only
```

Diagnose (BEFORE)

```
R3# show ip route 192.168.30.0      -> C 192.168.30.0/24 is directly connected, Loopback0 (route exists locally)
R2# show ip route eigrp            -> 192.168.30.0/24 is NOT in the table
R3# show ip protocols              -> shows "EIGRP-IPv4 Protocol ... Stub, receive-only"
R1# ping 192.168.30.1              -> Request timed out / unreachable
```

Fix and verify (AFTER)

```
R3(config)# router eigrp 100
R3(config-router)# eigrp stub connected summary

R2# show ip route eigrp
D    192.168.30.0/24 [90/130816] via 10.2.0.2, 00:00:12, GigabitEthernet0/2
R1# ping 192.168.30.1              -> !!!!! Success rate is 100 percent
```

Symptom D - eBGP session to the secondary ISP keeps resetting

What maximum-prefix is for: it protects your router from receiving too many routes (an ISP mistake, a route leak, or an attack) which could exhaust memory/CPU. **Why the session is torn down and not just ignored:** if the router quietly dropped extra routes, you would run with an incomplete, unpredictable routing table. Tearing the session down is a loud, safe failure. By default the session stays down (Idle) until cleared or until a restart timer expires, and may reset again if the neighbor still sends too many prefixes. This is the repeated reset you see.

Break

```
R1(config)# router bgp 65100
R1(config-router)# neighbor 172.16.2.2 maximum-prefix 3      ! ISP2 sends 5, limit is 3
```

Diagnose (BEFORE) - check the limit, find out why there are more routes

```
R1# show ip bgp summary
Neighbor      V      AS MsgRcvd MsgSent TblVer  InQ  OutQ Up/Down  State/PfxRcd
172.16.2.2    4  65002      11      13       1    0    0 00:00:05 Idle (PfxCt)

%BGP-3-MAXPFX: No. of prefix received from 172.16.2.2 (afi 0) reaches 4, max 3
%BGP-5-ADJCHANGE: neighbor 172.16.2.2 Down - Peer over limit
```

```
R1# show ip bgp neighbors 172.16.2.2 | include maximum|Maximum
Maximum prefixes allowed 3
```

```
ISP2# show ip bgp neighbors 172.16.2.1 advertised-routes      -> lists the 5 networks ISP2 sends
```

The configured limit is visible with `show ip bgp neighbors` or `show running-config | section bgp`. To see why more routes than expected, look at what the ISP advertises (as above) or `enable neighbor ... soft-reconfiguration inbound` and use `received-routes`.

Fix and verify (AFTER)

```
R1(config-router)# neighbor 172.16.2.2 maximum-prefix 10 80    ! limit 10, warn at 80 percent
R1# clear ip bgp 172.16.2.2
```

```
R1# show ip bgp summary
Neighbor      V      AS MsgRcvd MsgSent TblVer  InQ  OutQ Up/Down  State/PfxRcd
172.16.2.2    4  65002      8       9       6    0    0 00:01:10 5
R1# show ip route bgp          -> five B 198.51.10x.0/24 routes
```

Safe re-establishment means: first confirm the extra prefixes are legitimate (ask the ISP), then raise the limit with headroom (and a warning threshold), then clear the session. Do not just remove the limit. If Packet Tracer does not accept maximum-prefix on your router model, show the config as representative text and explain it in the report.

Symptom E - Burst of large fragmented ICMP echo (Wireshark)

What the pattern means: many ICMP echo requests, larger than the 1500-byte MTU so they are split into **fragments**, arriving very quickly from one source to a single host. That is a **ping flood / Ping-of-Death style DoS attempt**. **Likely intent:** exhaust the target's CPU, memory (fragment reassembly) or bandwidth so the service becomes slow or unavailable.

Reproduce

```
PC2> ping -l 10000 -n 200 192.168.20.100      ! large payload, many packets (use a smaller -l if PT refuses)
```

Wireshark is not part of Packet Tracer. Use **Simulation mode** (bottom right), filter on ICMP, send the ping, and screenshot the Event List and a PDU Information window; this is your 'capture'. In real Wireshark you would use

the filter below.

```
icmp.type == 8 && ip.flags.mf == 1          (echo requests with More Fragments set)
Representative capture (real Wireshark style):
No.   Time         Source            Destination      Proto Info
101   0.000100    192.168.10.66     192.168.20.100   IPv4 Fragmented IP protocol (id=0x2a1f, off=0)
102   0.000150    192.168.10.66     192.168.20.100   IPv4 Fragmented IP protocol (id=0x2a1f, off=1480)
103   0.000180    192.168.10.66     192.168.20.100   ICMP Echo (ping) request (reassembled)
... same source, same destination, hundreds in a few seconds
```

Two concrete mitigations

1. ACL (works in Packet Tracer). Block the attacker, allow normal ping replies, apply inbound on the WAN-facing interface of R2:

```
R2(config)# ip access-list extended PROTECT-FAB
R2(config-ext-nacl)# deny icmp host 192.168.10.66 host 192.168.20.100 echo
R2(config-ext-nacl)# permit icmp any any echo-reply
R2(config-ext-nacl)# permit ip any any
R2(config)# interface gi0/0
R2(config-if)# ip access-group PROTECT-FAB in

PC2> ping 192.168.20.100          -> Request timed out
PC1> ping 192.168.20.100          -> still works (PC1 is not blocked)
R2# show access-lists
Extended IP access list PROTECT-FAB
 10 deny icmp host 192.168.10.66 host 192.168.20.100 echo (45 matches)
```

2. Rate-limit ICMP (real IOS; usually not available in Packet Tracer - show as representative config):

```
access-list 120 permit icmp any any echo
class-map match-all ICMP-ECHO
 match access-group 120
policy-map LIMIT-ICMP
 class ICMP-ECHO
  police 64000 conform-action transmit exceed-action drop
interface gi0/0
 service-policy input LIMIT-ICMP
```

Other helpers: filter or drop fragments to the server, block ICMP at the edge from untrusted sources, and set the server host firewall.

5. Are the five symptoms related? (Objective 6)

Short answer: more than one unrelated fault. Symptoms A and C may come from one maintenance window at the Fab area, but B, D and E are independent.

Narrative (about 400 words)

Taken together, the five symptoms are best explained by more than one unrelated fault rather than a single trigger. Each lives in a different protocol and layer: A is a Layer 2 spanning-tree event, B is an OSPF adjacency problem on the WAN, C is an EIGRP route-filtering decision, D is an eBGP session control at the Internet edge, and E is hostile traffic. A single configuration mistake normally leaves a common fingerprint, for example every protocol on one router failing at the same time, and the symptom log shows nothing like that.

The devices also differ. B involves the HQ and Fab routers across the multipoint WAN, D happens only on the HQ Internet edge, and C is on the Test and Packaging router. A, C and E all sit in the Fab and Test and Packaging area, and a recent change there could plausibly explain A, where a switch priority was 'recently' lowered, and C, where a router was 'recently' configured as a stub. Those two are linked by timing and by the word 'recently', so they may come from one change ticket, but they are still two separate misconfigurations with two separate fixes. Even then, one change cannot explain D, because the BGP edge is at HQ, or E, because a flood is generated by a host and not by a router setting.

Several symptoms also follow a 'cause, then protective reaction' pattern. Root Guard did what it is designed to do, the maximum-prefix limit did what it is designed to do, and the stub did what its type tells it to do. The protections worked; the root causes are upstream: a wrongly lowered priority, too many prefixes for the configured limit, a wrong stub type. Symptom B is different because the missing or mismatched NBMA configuration is itself the fault, and symptom E is the only one with an external actor.

My conclusion is to review the change log for A and C as a possible single change window, and to treat B, D and E as independent faults. Each must be fixed and verified on its own, because repairing one will not repair the others. To prevent repeats, use change control, Root Guard plus BPDU Guard on edge ports, prefix limits with headroom and warnings, a stub-design review, and ICMP rate limiting and ACLs at the Fab boundary.

6. Deliverable 1 - Root-cause analysis table

Symptom	Likely cause	Evidence	Fix	Verification
A Root Guard	SW4 priority lowered to 0, sends superior BPDU	Log ROOTGUARD_BLOCK; inconsistentports shows fa0/3 Root Inconsistent	Set SW4 priority back to 32768	inconsistentports = 0; ROOTGUARD_UNBLOCK log
B OSPF ATTEMPT	R2 side not set to non-broadcast (hub neighbor not answered / statement missing)	show ip ospf neighbor: ATTEMPT for 10.0.0.2, R3 FULL; network types differ	Set non-broadcast on R2; ensure neighbor 10.0.0.2 on R1	show ip ospf neighbor: R2 and R3 FULL
C EIGRP stub	R3 uses eigrp stub receive-only	192.168.30.0 in R3 table but not on R2; show ip protocols shows receive-only	eigrp stub connected summary	R2 has D 192.168.30.0/24; ping works
D BGP prefix limit	maximum-prefix 3 but ISP2 sends 5	MAXPFX log; state Idle (PfxCt); limit seen in neighbor output	Raise limit with warning level, verify prefixes, clear ip bgp	Session Established, PfxRcd 5
E ICMP flood	Large fragmented echo flood from 192.168.10.66 to 192.168.20.100	Capture shows MF fragments, same src/dst, rapid burst	ACL on R2 and ICMP rate-limit	ACL match counters; PC1 still pings; flood blocked

7. Deliverable 3 - Prioritized remediation plan

Order	Symptom	Business impact	Why this order	Verification command
1	B OSPF WAN link	HQ to Fab connectivity on the backbone is degraded; production systems and ERP depend on it	The WAN is the backbone; other sites and fixes rely on it	show ip ospf neighbor (all FULL)
2	C EIGRP stub	Test & Packaging network unreachable from the rest of the company; test results cannot reach HQ	Whole site is cut off from some paths	show ip route eigrp; ping 192.168.30.1
3	A Root Guard	Devices behind the blocked port at the Fab lose network; STP security risk	One access port, local impact; quick fix	show spanning-tree inconsistentports
4	E ICMP flood	Possible slowdown of a Fab host; availability risk but no outage yet	Contain quickly: the ACL takes minutes and can be done in parallel	show access-lists (match counters)
5	D BGP secondary	Only the backup Internet path is flapping; primary ISP still carries traffic	Redundancy lost, not an outage, so last	show ip bgp summary (Established)

Quick win note: the E ACL takes about 2 minutes and can be applied first as containment while B, C and A are fixed.

8. Deliverable 4 - Before / after evidence

Use the BEFORE and AFTER outputs shown in section 4 (they are representative). In your own submission replace them with **your own Packet Tracer screenshots**. A simple layout for each symptom: one screenshot of the BEFORE command, one line saying what is wrong, one screenshot of the AFTER command, one line saying it is fixed.

9. Simple way to complete the project

#	Task	Time
1	Build the topology (Figure 1) and name the devices	20 min
2	Do basic IPs, then the healthy base configuration (Step 3)	30 min
3	Run the Step 4 checks and screenshot the healthy baseline	5 min

#	Task	Time
4	For A to E: break -> BEFORE screenshot -> fix -> AFTER screenshot (do one at a time and fix each before breaking the next)	60 min
5	Copy the root-cause table (section 6) and replace with your results	10 min
6	Copy or adapt the narrative (section 5), about 400 words, in your own words	10 min
7	Copy the remediation table (section 7)	5 min
8	Put everything in one report: topology screenshot, 5 BEFORE/AFTER pairs, 4 deliverables. Save the .pkt file	10 min

Packet Tracer limits (be honest in your report)

Item	What to do
Wireshark not available	Use Simulation mode and PDU details as the capture; mention the Wireshark filter as representative
ATTEMPT state may not show	Screenshot what you see (Down/Init) and explain ATTEMPT
ICMP rate-limiting (police)	Show the config as representative text; the ACL is the lab-verified mitigation
Root Guard / maximum-prefix commands rejected	Try another router/switch model (2911, 2960); if still rejected, present the commands as representative
Large ping size refused	Reduce -l (for example 4000) so it still fragments (over 1472 bytes)

Tips

- Break and fix **one symptom at a time**. Save the .pkt file after the healthy baseline.
- Always run `copy running-config startup-config` if you want the config kept.
- If something stops working unexpectedly, run `show ip interface brief` first and check for shut interfaces.